

For the theorem's packing assertion, let $U \in P$ be given. For all $j = 1, \dots, k$ let S_j be the subgraph of T_j induced on U by its edges in D . These forests S_j are edge-disjoint, since by the maximality of $E(\mathcal{T})$ and Lemma 2.4.5 no edge in D lies in more than one T_j . Let us show that the S_j are connected.

Since the edges from D form a connected submultigraph on U , it suffices to show that for every edge $uu' \in D$ with $u, u' \in U$ there is a u - u' path in S_j . This is clear if uu' lies in T_j , and hence in S_j . If it does not, then the path uT_ju' still has all its edges e in D , and hence lies in S_j : if $e_0, \dots, e_n$ is an exchange chain witnessing that $e_0 = uu' \in D$, then $e, e_0, \dots, e_n$ is an exchange chain putting e in D , because e lies on the fundamental cycle of e_0 with respect to T_j .

As every T_j induces connected subgraphs S_j on the partition classes of P , contracting these S_j turns the T_j into spanning trees T'_j of G/P . These T'_j cover all the edges of G/P , since $E \setminus E(\mathcal{T}) \subseteq D$. $\square$

The packing-covering theorem differs from both the tree packing and the tree covering theorem in a fundamental way. The non-trivial directions of the latter two theorems each obtain a structural assertion about a graph, the existence of a packing or covering, as a consequence of quantitative assumptions about all their minors of a certain type: contraction minors for the packing theorem, and 'deletion minors' – i.e., subgraphs – for the covering theorem. This format makes them interesting: they offer valuable structural information for one graph in exchange for less valuable quantitative information about many smaller graphs.

The packing-covering theorem, by contrast, makes a structural assertion about every graph: with no need for any assumptions at all, neither quantitative nor qualitative.

The packing-covering theorem extends to infinite graphs in two interestingly different ways; see Exercises 20 and 131 in Chapter 8.

2.5 Path covers

Let us return once more to König's duality theorem for bipartite graphs, Theorem 2.1.1. If we orient every edge of G from A to B , the theorem tells us how many disjoint directed paths we need in order to cover all the vertices of G : every directed path has length 0 or 1, and clearly the number of paths in such a 'path cover' is smallest when it contains as many paths of length 1 as possible – in other words, when it contains a maximum-cardinality matching.

In this section we put the above question more generally: how many paths in a given directed graph will suffice to cover its entire vertex set? Of course, this could be asked just as well for undirected graphs. As it turns out, however, the result we shall prove is rather more trivial in

the undirected case (exercise), and the directed case will also have an interesting corollary.

A *directed path* is a directed graph $P \neq \emptyset$ with distinct vertices $x_0, \dots, x_k$ and edges $e_0, \dots, e_{k-1}$ such that e_i is an edge directed from x_i to x_{i+1} , for all $i < k$. In this section, *path* will always mean ‘directed path’. The vertex x_k above is the *last vertex* of the path P , and when $\mathcal{P}$ is a set of paths we write $\text{ter}(\mathcal{P})$ for the set of their last vertices. A *path cover* of a directed graph G is a set of disjoint paths in G which together contain all the vertices of G .

path

 $\text{ter}(\mathcal{P})$

path cover

Theorem 2.5.1. (Gallai & Milgram 1960)

Every directed graph G has a path cover $\mathcal{P}$ and an independent set $\{v_P \mid P \in \mathcal{P}\}$ of vertices such that $v_P \in P$ for every $P \in \mathcal{P}$.

Proof. Clearly, G has a path cover, e.g. by trivial paths. We prove by induction on $|G|$ that for every path cover $\mathcal{P} = \{P_1, \dots, P_m\}$ with $\text{ter}(\mathcal{P})$ minimal there is a set $\{v_P \mid P \in \mathcal{P}\}$ as claimed. For each i , let v_i denote the last vertex of P_i .

 $\mathcal{P}, P_i$ v_i

If $\text{ter}(\mathcal{P}) = \{v_1, \dots, v_m\}$ is independent there is nothing more to show, so we assume that G has an edge from v_2 to v_1 . Since $P_2 v_2 v_1$ is again a path, the minimality of $\text{ter}(\mathcal{P})$ implies that v_1 is not the only vertex of P_1 ; let v be the vertex preceding v_1 on P_1 . Then $\mathcal{P}' := \{P_1 v, P_2, \dots, P_m\}$ is a path cover of $G' := G - v_1$ (Fig. 2.5.1). Clearly, any independent set of representatives for $\mathcal{P}'$ in G' will also work for $\mathcal{P}$ in G , so all we have to check is that we may apply the induction hypothesis to $\mathcal{P}'$. It thus remains to show that $\text{ter}(\mathcal{P}') = \{v, v_2, \dots, v_m\}$ is minimal among the sets of last vertices of path covers of G' .

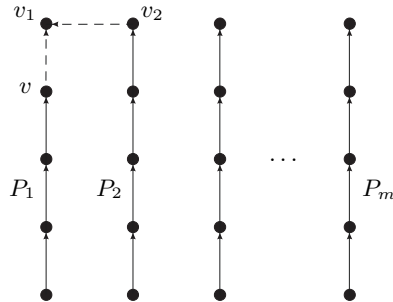
 v $\mathcal{P}', G'$ 

Fig. 2.5.1. Path covers of G and G'

Suppose then that G' has a path cover $\mathcal{P}''$ with $\text{ter}(\mathcal{P}'') \subsetneq \text{ter}(\mathcal{P}')$. If a path $P \in \mathcal{P}''$ ends in v , we may replace P in $\mathcal{P}''$ by Pvv_1 to obtain a path cover of G whose set of last vertices is a proper subset of $\text{ter}(\mathcal{P})$, contradicting the choice of $\mathcal{P}$. If a path $P \in \mathcal{P}''$ ends in v_2 (but none in v), we similarly replace P in $\mathcal{P}''$ by Pv_2v_1 to obtain a contradiction to the

minimality of $\text{ter}(\mathcal{P})$. Hence $\text{ter}(\mathcal{P}'') \subseteq \{v_3, \dots, v_m\}$. But now $\mathcal{P}''$ and the trivial path $\{v_1\}$ together form a path cover of G that contradicts the minimality of $\text{ter}(\mathcal{P})$. $\square$

As a corollary to Theorem 2.5.1 we obtain a classical result from the theory of partial orders. Recall that a subset of a partially ordered set $(P, \leq)$ is a *chain* in P if its elements are pairwise comparable; it is an *antichain* if they are pairwise incomparable.

chain
antichain

Corollary 2.5.2. (Dilworth 1950)

In every finite partially ordered set $(P, \leq)$, the minimum number of chains with union P is equal to the maximum cardinality of an antichain in P .

Proof. If A is an antichain in P of maximum cardinality, then clearly P cannot be covered by fewer than $|A|$ chains. The fact that $|A|$ chains will suffice follows from Theorem 2.5.1 applied to the directed graph on P with the edge set $\{(x, y) \mid x < y\}$. $\square$

Exercises

- Let M be a matching in a bipartite graph G . Show that if M is sub-optimal, i.e. contains fewer edges than some other matching in G , then G contains an augmenting path with respect to M . Does this fact generalize to matchings in non-bipartite graphs?
- (continued)
Describe an algorithm that finds, as efficiently as possible, a matching of maximum cardinality in any bipartite graph.
- Show that if there exist injective functions $A \rightarrow B$ and $B \rightarrow A$ between two infinite sets A and B then there exists a bijection $A \rightarrow B$.
- Moving alternately, two players jointly construct a path in some fixed graph G . If $v_1 \dots v_n$ is the path constructed so far, the player to move next has to find a vertex v_{n+1} such that $v_1 \dots v_{n+1}$ is again a path. Whichever player cannot move loses. For which graphs G does the first player have a winning strategy, for which the second?
- Derive the marriage theorem from König's theorem.
- Let G and H be defined as for the third proof of Hall's theorem. Show that $d_H(b) \leq 1$ for every $b \in B$, and deduce the marriage theorem.
- Does our first proof of the marriage theorem use the assumption that the graph is finite? If so, can it be adapted so that it works for infinite graphs too?